

6 (long) Consider the air-standard Stirling cycle, which is the same as the Otto cycle *except* that compression and expansion occur isothermally rather than adiabatically.

(a) Sketch the cycle on a pressure - volume diagram and label the following points: 1 – beginning of compression; 2 – end of compression; 3 – beginning of expansion; 4 – end of expansion. Label the heat transfer as positive (in) or negative (out) for each process. [5]

(b) For a Stirling engine cycle using a perfect gas and a fixed volumetric compression ratio r_v , operating between two reservoirs of high T_h , and low T_l , temperatures, show that the inward specific heating for the cycle is $q_{in} = k_1 T_h - k_2 T_l$. Find k_1 and k_2 in terms of r_v , specific heat capacity c_v , and gas constant R . [5]

(c) Consider a Stirling engine cycle operating with air (taken as a perfect gas) where the state at the beginning of the isothermal compression process is $p_1 = 100$ kPa, $T_l = 300$ K, $r_v = 6$, and the maximum temperature in the cycle is $T_h = 1400$ K.

(i) Calculate the thermal efficiency of the cycle. Assume that no heat is recycled. [6]

(ii) Calculate the thermal efficiency of the cycle assuming an ideal regenerator, where the heat added from point 2 to 3 is supplied by the rejected heat from 4 to 1. [6]

(d) Determine the heat transferred into the engine and thermal efficiency of the engine, treating the air in the engine as a semi-perfect gas where the specific heat capacity is given as $c_v = \alpha + \beta T + \gamma T^2$, where $\alpha = 610$ J kg⁻¹ K⁻¹, $\beta = 0.3$ J kg⁻¹ K⁻², $\gamma = 7 \times 10^{-5}$ J kg⁻¹ K⁻³. Assume no heat is recycled. [8]